Subsonic High Reynolds Number: SD8020 airfoil

Case Details

The objective of the simulation is to validate the SD8020 airfoil's experimental data given in [1] with low computational cost for fast but accurate results.

- Reference: `https://m-selig.ae.illinois.edu/uiuc_lsat/Low-Speed-Airfoil-Data-V1.pdf`
- Airfoil: SD8020 airfoil
- Mach Number: 0.15
- Reynolds Number: $100,000$
- Angle of Attack: 4°

How to Run

1. Select SD8020 airfoil from the airfoil directory
2. Enter the desired number of coordinates if required and output directory.
3. Click on "Analyze" to obtain the accurate parameterization method and its respective parameters
4. Go to SU2 Settings → Meshing & Conditions
5. Enter Mach, Reynolds number and enter 4 in minimum AoA with 0 step.
6. Select "Load recommended settings"
7. Choose the type of mesh and target y+ value
8. Select "Generate Structured Mesh"
9. Go to SU2 Settings → Solver Setup & Run
10. Modify settings as desired or select "Run SU2 Analysis" (validated settings have been provided as default and are given here in the next section)
11. After convergence, the output or post-processed contours will be displayed in the SU2 Settings → Results tab and saved in the output directory.

Solver Settings

Given below are the solver settings validated for this case and are provided by default for this regime. This case is run using unsteady parameters:

Setting	**File (HighRe.cfg)**
NUM_METHOD_GRAD	WEIGHTED_LEAST_SQUARES
CONV_NUM_METHOD_FLOW	JST
SLOPE_LIMITER_FLOW	VENKATAKRISHNAN
LINEAR_SOLVER	FGMRES
LINEAR_SOLVER_PREC	ILU
KIND_TURB_MODEL	SST
KIND_TRANS_MODEL	LM (NONE for Re¿100,000
CFL_NUMBER	10
CONV_FIELD	RMS_PRESSURE
Min log10 Residual	-8
TIME STEPPING	NO

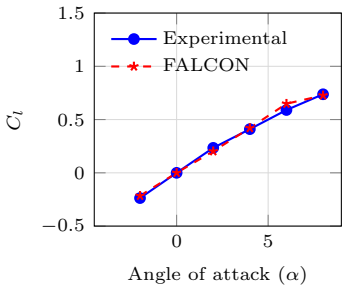
Table 1: Settings used in `HighRe.cfg`

Result

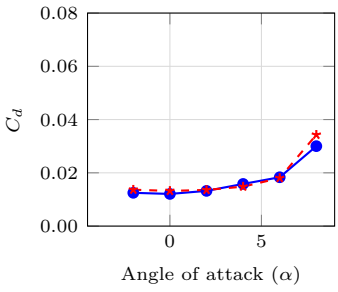
AoA	Cl_{exp}	Cd_{exp}	Cl_{obt}	Cd_{obt}	**Cl Error (%)**	**Cd Error (%)**
4	0.4101	0.0158	0.4205	0.014818	2.53	6.39

Table 2: Results at Mach 0.15 with comparison between experimental and obtained data

Figure **??** shows the results over angle of attack sweep from -2 to 4 degrees.



(a) SD 8020, C_l



(b) SD 8020, C_d

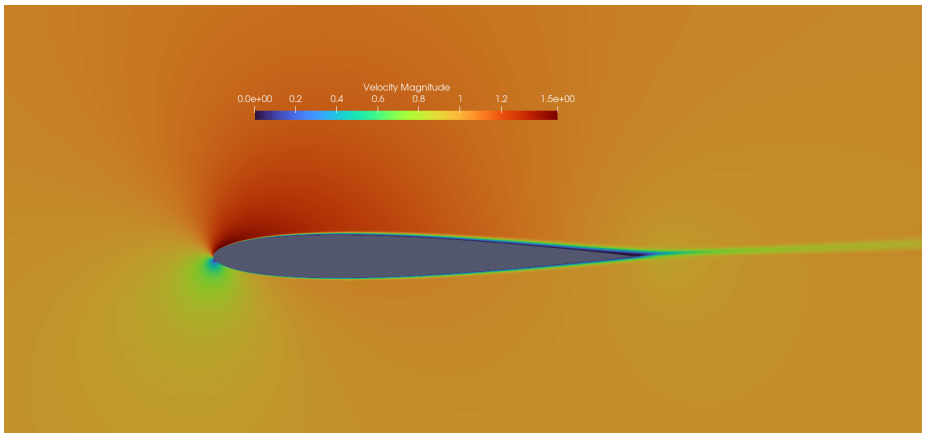


Figure 2: Velocity field as visualized in ParaView

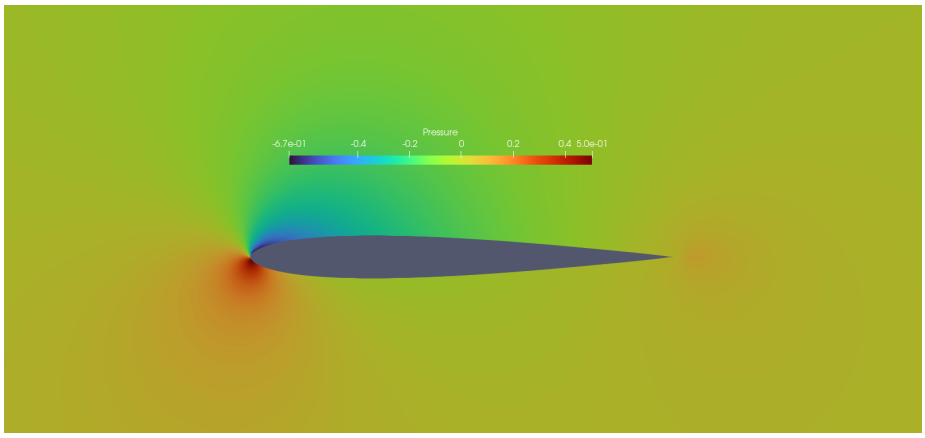


Figure 3: Pressure field as visualized in ParaView

References

[1] M. S. Selig, *UIUC Airfoil Data Site*, Department of Aerospace Engineering, University of Illinois at Urbana-Champaign, 1996. Available online at `https://m-selig.ae.illinois.edu/ads.html`.